Streamlit for Web Application and UI Development

1. Introduction

Streamlit is an open-source Python framework designed to build interactive web applications for data science and machine learning projects. It allows developers to create web-based user interfaces directly from Python scripts without requiring extensive knowledge of front-end technologies such as HTML, CSS, or JavaScript.

Streamlit is widely used for **rapid prototyping, visualization dashboards, machine learning applications, and AI-powered tools such as RAG-based chatbots**. Its simplicity and integration with popular Python libraries make it a preferred framework for academic research and capstone projects.

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2. Key Features of Streamlit

2.1 Simple Python-Based Development

Streamlit applications are written entirely in Python. Developers can convert a Python script into a web application using a single command:

streamlit run app.py

This approach eliminates the need for traditional web frameworks and significantly reduces development time.

2.2 Interactive User Interface Components

Streamlit provides built-in UI widgets such as:

- Text input boxes
- Dropdown menus
- Sliders
- Buttons
- File uploaders
- Chat interfaces
- Data tables

These widgets enable users to interact with machine learning models and data processing pipelines easily.

2.3 Real-Time Data Visualization

Streamlit integrates seamlessly with visualization libraries such as:

- Matplotlib
- Plotly
- Altair

- Pandas

This allows developers to display charts, graphs, and dashboards dynamically within the web application.

2.4 Rapid Prototyping

Streamlit enables quick development and testing of machine learning workflows. Changes in code are automatically reflected in the application through **live reloading**, making experimentation faster.

2.5 Integration with AI and ML Frameworks

Streamlit works efficiently with modern AI frameworks such as:

- LangChain
- Ollama
- Hugging Face Transformers

This makes it suitable for developing **AI-powered applications such as document search tools, medical research assistants, and RAG-based question-answering systems**.

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3. Architecture of a Streamlit-Based Application

A typical Streamlit web application consists of the following components:

1. User Interface Layer

This layer handles user interaction using Streamlit widgets such as input boxes, file uploaders, and buttons.

2. Application Logic Layer

This layer processes user inputs and executes Python functions such as:

- Data preprocessing
- Model inference
- Document retrieval
- Query generation

3. Backend AI/ML Components

This layer integrates machine learning or AI models such as:

- Retrieval systems
- Vector databases
- Large Language Models (LLMs)

4. Output Visualization Layer

Results are displayed in the Streamlit interface through:

- Text responses
- Tables
- Charts
- Interactive components

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4. Role of Streamlit in RAG-Based Chatbots

In Retrieval-Augmented Generation (RAG) systems, Streamlit is commonly used as the **frontend interface** for interacting with the AI system.

The typical workflow includes:

1. User enters a query in the Streamlit interface.
2. The query is sent to a retrieval system.
3. Relevant documents are retrieved from a database.
4. A large language model generates an answer using the retrieved context.
5. The response is displayed in the Streamlit web interface.

This architecture allows researchers to create **interactive AI research assistants for literature review and knowledge discovery**.

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5. Advantages of Using Streamlit

- Easy to learn and implement
- Minimal frontend development required
- Rapid deployment of ML applications
- Built-in support for interactive widgets
- Strong integration with Python ecosystem
- Ideal for research prototypes and capstone projects

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6. Limitations of Streamlit

Despite its advantages, Streamlit has some limitations:

- Limited customization compared to full-stack frameworks
- Not ideal for very large-scale production applications
- Performance can decrease with extremely complex workflows
- Limited control over backend infrastructure

However, for **academic research, prototypes, and AI demonstration systems**, Streamlit remains highly effective.

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7. Relevance to the Capstone Project

In this capstone project, Streamlit is used to develop the **user interface of the AI-driven research assistant system**. The framework enables users to:

- Upload biomedical research documents
- Submit research-related queries
- Interact with the AI-powered question-answering system
- Visualize retrieved evidence and generated insights

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8. Streamlit and RAG chatbot integration diagram